

The residual, bounded at every instant

drop-in for Sec. VIII; the RMS form is (VIII.3)

0.1 What is to be proved, and what is not true

Write the recorded rate over the post-onset window as the nominal response, the deviation the model does not carry, and sensor noise,

$$y(\tau) = \omega_{\text{nom}}(\tau) + e_\omega(\tau) + n(\tau), \quad (\text{P.1})$$

and let $\hat{\omega}$ be the fitted curve, chosen to minimise $\|y - g\|^2$ over the model family. The natural statement to want about the residual $r = y - \hat{\omega}$ is

$$|r(\tau)| \stackrel{?}{\leq} |e_\omega(\tau)| + k\sigma_n. \quad (\text{P.2})$$

It is false, and the counterexample is at the onset. The envelope of (93) vanishes there, $E(\tau) \propto \sinh C_2\tau \rightarrow 0$, and so does e_ω ; the residual does not. Minimisation is over the whole window, so a deviation living at large τ moves the fitted curve at $\tau = 0$ as well, and (P.2) would require that movement to vanish where e_ω does. No multiplicative form $|r| \leq \Gamma|e_\omega| + k\sigma_n$ can hold either, for the same reason: measured on the campaign the ratio of the true bound to E is 2.1 at the end of the window and 27 to 95 at the first sample.

What is missing is the displacement of the fit itself. It costs one term, and — this is the point of the section — that term is expressible in e_ω and n alone, so the statement stays in the form (P.2) asks for.

0.2 The proof

Three steps. The first two are where the work is and neither needs more than (VIII.3) already assumes.

Step 1: optimality bounds the displacement, in norm. The nominal is itself a member of the family — it is a hyperbolic cosine with the true amplitude and exponent — so the minimiser cannot be farther from the data than it is. With the triangle inequality,

$$\|\hat{\omega} - \omega_{\text{nom}}\| \leq \|\hat{\omega} - y\| + \|y - \omega_{\text{nom}}\| \leq 2\|e_\omega + n\|, \quad (\text{P.3})$$

the second step using $\|\hat{\omega} - y\| = \min_g \|g - y\| \leq \|\omega_{\text{nom}} - y\|$. Nothing here is linear: (P.3) holds for any family containing ω_{nom} and any global minimiser over it.

When the family *is* linear the factor drops to one. The estimator holds C_2 at the value fixed by the rig and fits the amplitude and the pre-onset level, so on the sample grid $\tau_1, \dots, \tau_N$ the admissible curves form a two-dimensional subspace

$$V = \text{span}\{u, \mathbf{1}\} = \text{range } A, \quad A = \begin{bmatrix} u & \mathbf{1} \end{bmatrix} \in \mathbb{R}^{N \times 2}, \quad u_i = \cosh C_2\tau_i - 1, \quad (\text{P.4})$$

the minimiser is $\hat{\omega} = Py$ with $P = A(A^\top A)^{-1}A^\top$, and since $\omega_{\text{nom}} \in V$ is fixed by P ,

$$\hat{\omega} - \omega_{\text{nom}} = P(e_\omega + n), \quad \|\hat{\omega} - \omega_{\text{nom}}\| \leq \|e_\omega + n\|. \quad (\text{P.5})$$

Step 2: in a finite-dimensional space, a norm bounds a value. The displacement lies in V , and for $v \in V$ we have $v = Pv$, so $v_i = \langle P_i, v \rangle$ and, since $\|P_i\|^2 = (P^2)_{ii} = P_{ii}$,

$$|v_i| \leq \sqrt{P_{ii}} \|v\|. \quad (\text{P.6})$$

This is exact, not an estimate, and $\sqrt{P_{ii}}$ is computable before a run is flown. It is the classical leverage of sample i ; over the campaign $\max_i \sqrt{P_{ii}} = 0.35$ to 0.40 .

Step 3: the triangle inequality. With $r = (\omega_{\text{nom}} - \hat{\omega}) + e_\omega + n$ and (P.5)–(P.6),

$$\boxed{|r_i| \leq |e_{\omega,i}| + \sqrt{P_{ii}} \|e_\omega\| + |n_i| + \sqrt{P_{ii}} \|n\|} \quad (\text{P.7})$$

Every term is e_ω or n . The deviation enters twice, once pointwise and once in norm, and the second copy is the displacement of the fit — which is exactly what (P.2) was missing and why it fails at the onset, where $|e_\omega(\tau)|$ vanishes and $\|e_\omega\|$ does not.

The noise term, sharpened. Bounding $\|Pn\|$ by $\|n\|$ throws away the only thing that makes a projector harmless. For $n \sim \mathcal{N}(0, \sigma_n^2 I)$ and P symmetric idempotent, $\mathbb{E}\|Pn\|^2 = \sigma_n^2 \text{tr } P = 2\sigma_n^2$ regardless of N , and $\text{Var}[(Pn)_i] = \sigma_n^2 P_{ii}$, so the last two terms of (P.7) are together at most $k\sigma_n(1 + \sqrt{P_{ii}})$ at confidence $2\Phi(k) - 1$, against the $\sigma_n\sqrt{N}$ that $\|n\|$ would give — a factor of 4.5 at $N = 41$. The form used below is therefore

$$|r_i| \leq \underbrace{E_i}_{\text{pointwise}} + \underbrace{\sqrt{P_{ii}}\|E\|}_{\text{displacement}} + k\sigma_n(1 + \sqrt{P_{ii}}), \quad (\text{P.8})$$

using $|e_{\omega,i}| \leq E_i$ and $\|e_{\omega}\| \leq \|E\|$ from (93). Two Gaussian events enter per sample, so a union bound over the window replaces k by $\Phi^{-1}(1 - \alpha/4N)$: for $N = 53$ that is 3.50 at 95% and 3.90 at 99%, against 1.96 and 2.58 for one sample.

0.3 Why this route and not the band

The displacement can also be bounded by carrying the band into the projector's kernel, $|(Pe_{\omega})_i| \leq \sum_j |P_{ij}|E_j$, which is sharp over the box $\{|e_{\omega,j}| \leq E_j\}$. That was the earlier form of this section and it is worth saying what replacing it buys.

Weaker hypotheses. The box argument needs the band pointwise at every sample. (P.7) needs $\|e_{\omega}\| \leq \|E\|$, one number, and otherwise only what (VIII.3) already assumes: ω_{nom} in the family, and a global minimum.

It survives freeing C_2 . Step 1 in the form (P.3) is projection-free, so with C_2 fitted — as the calibration stage does — the displacement is still bounded, by $2\|e_{\omega} + n\|$. Only step 2 needs replacing, by the corresponding inequality on the two-dimensional manifold. The curvature (reach) correction that the box route needed throughout is confined to that one step.

It measures tighter. Applied to every sample of all 140 runs the box form leaves a worst ratio of 0.763 and (P.8) leaves 0.611, because $\sqrt{P_{ii}}$ falls near the onset while $\sum_j |P_{ij}|E_j$ does not.

0.4 The constants, and why they do not grow

Everything in (P.8) is computable before a run is flown: P depends only on C_2 and the sample grid, E only on the geometry box.

$\dot{M}$ N m/s	N	$\text{tr } P$	$\kappa = \max_i \sqrt{P_{ii}}$	$\ E\ /\sup E$	$\kappa\ E\ /\sup E$
0.10	74	2.000	0.352	3.13	1.12
0.20	63	2.000	0.362	3.12	1.15
0.30	57	2.000	0.370	3.10	1.17
0.45	54	2.000	0.376	3.09	1.18
0.65	48	2.000	0.385	3.07	1.21
0.90	44	2.000	0.395	3.02	1.22
1.20	41	2.000	0.404	2.99	1.23

The trace is 2 at every rate and every window length, being the dimension of the range, and that is why the constants do not grow with the record: a two-dimensional projector cannot move much, however many samples it is given. The displacement term is 1.09 to 1.29 times $\sup E$ across all 140 runs, so in uniform form

$$\max_i |r_i| \lesssim 2.3 \sup_{\tau} E(\tau) + 1.4 k\sigma_n.$$

P itself is the standard least-squares projector: minimising $\|y - A\theta\|^2$ gives the normal equations $A^{\top}A\hat{\theta} = A^{\top}y$, whose content is $A^{\top}(y - A\hat{\theta}) = 0$ — the residual is orthogonal to everything the model can produce. For these two columns it collapses to a form with no matrix inverse left in it,

$$P_{ij} = \frac{1}{N} + \frac{\tilde{u}_i \tilde{u}_j}{\sum_k \tilde{u}_k^2}, \quad \tilde{u} = u - \bar{u}, \quad (\text{P.9})$$

the constant channel plus the u channel with the constant removed, reproducing the numerically formed P to 10^{-16} . Every property used above is one line from it: $P^2 = P$ and $P^{\top} = P$ are immediate, $Pv = v$ on V gives (P.5), $\text{Var}[(Pn)_i] = \sigma_n^2 P_{ii}$ gives the noise term, and $\text{tr } P = 1 + 1 = 2$. It also locates the worst sample: P_{ii} is largest where $|\tilde{u}_i|$ is, which is the last sample of the window in all 140 runs, and only $1/N = 0.013$ to 0.024 of P_{ii} there is the constant — the rest is the amplitude direction, whose cosh has grown by then.

0.5 The bound tested, and the two things it exposes

Applied to every sample of every run at $k = 3.5$:

$\dot{M}$ N m/s	runs entirely inside σ_{pre}	inside σ_{in}	worst ratio	deployed worst	$\sigma_{\text{pre}} / \sigma_{\text{in}}$ °/s
0.10	20/20	20/20	0.324	0.369	0.087 / 0.831
0.20	20/20	20/20	0.391	0.551	0.044 / 0.800
0.30	18/20	20/20	0.457	0.584	0.101 / 0.823
0.45	19/20	20/20	0.491	0.979	0.127 / 0.707
0.65	18/20	20/20	0.580	0.863	0.122 / 0.856
0.90	15/20	20/20	0.556	1.030	0.111 / 0.749
1.20	13/20	20/20	0.611	1.212	0.173 / 0.773

The bound holds on all 140 runs at every sample, the worst reaching 0.611 of it. Two conditions on how it is applied are not optional, and the table makes both visible.

The noise scale must be measured in the window. Taking σ_n from the pre-onset record leaves 123 of 140 inside, and the shortfall is at the fast end. That is not the theorem failing but its hypothesis: before the onset the vehicle is carried by every landing gear, after it pivots on one edge, and the in-window disturbance is 0.71 to 0.86°/s against a pre-onset floor of 0.04 to 0.17. The honest estimate is the in-window content above 5 Hz, which the fitted model provably cannot produce — $\hat{\omega}$ is smooth on the scale of the window — so whatever is there is disturbance by construction rather than by assumption.

A pinned amplitude adds a term. V is the family the bound is about: amplitude and level free. The shipped estimator frees neither. It takes $C_1 = K\dot{M}$ from the calibration stage, and it forms the level C from the *pre-onset* segment rather than minimising over it on the window, so once the onset is chosen its post-onset curve has no free parameter at all and $\hat{\omega} \neq Py$. Step 1 then removes only part of the nominal and two terms survive,

$$|r_i| \leq |\Delta C_1| u_i + |C - b| + E_i + \sqrt{P_{ii}} \|E\| + k\sigma_n(1 + \sqrt{P_{ii}}), \quad (\text{P.8}')$$

the first being the calibration’s amplitude error and the second the baseline channel of (105)–(106). Their measured sizes are very different. The amplitude term is $|\Delta C_1|/C_1 = 11.3\%$ and it is what puts 26 runs outside the RMS form (VIII.3); without it the deployed residual reaches 1.21 of the bound. The baseline term is 0.005 to 0.029°/s, under 2% of $\sup E$. One constraint matters and the other does not, and the bound says which.

0.6 What the calibration is worth: step 1 needs C_2 exact

Step 1 removes the nominal because it lies in the family, and the family is built from the calibrated C_2 . If that value is not the true one the nominal leaves V and the cancellation is only partial: with $C_2 = C_2^*(1 + \varepsilon_2)$ the residual retains $(I - P)\omega_{\text{nom}}$, and (P.8) gains a further term. This is the price of conditioning on the calibration stage — which fits C_2 freely, by a per-run nonlinear least squares and a per-configuration median — and it is small, almost linear in ε_2 :

$\dot{M}$ N m/s	$\varepsilon_2 = 1\%$	$\varepsilon_2 = 2\%$	$\varepsilon_2 = 5\%$	$\varepsilon_2 = 10\%$
0.10	0.005	0.010	0.027	0.059
0.45	0.013	0.027	0.071	0.152
0.90	0.018	0.036	0.093	0.195
1.20	0.020	0.041	0.105	0.221

A 2% exponent error costs 1 to 4% of the envelope and a 5% error costs 3 to 11%, the fast ramp being the more exposed because its window is shorter and the two hyperbolic shapes are less separated there. Like the pinned amplitude it enters additively and is not covered by $\bar{\rho}$, which bounds the physics the model omits and says nothing about the constants the model is given.

The empirical check above is unaffected by any of this: it compares a *measured* record, which contains whatever exponent the rig actually has, against the envelope, so the 140/140 already includes the misspecification. What the term supplies is the proof that this had to be so, and a budget line for it.